

VapourWiki — Audion VS Engine preset reference (EN)

Full English reference for all 28 presets across 4 palettes. Starts with a **decision tree** ("which preset for which job"), then per-preset detail: what it does, what material it suits, key parameters, recommended encoder.

Russian counterpart: `VapourWiki_RU.md` (this document mirrors it). Per-preset technical docstrings live inside each `.vpy` file in `system_core/presets/<palette>/`.

Before you start: install the engine

The build ships without VapourSynth, its plugins and the vs-mlrt models — some seventy modules, each under its own licence, close to three gigabytes together. The program installs them itself, from their authors.

Until that is done, not one of the presets described below will run.

Run `builder_main.cmd` and pick, in order: `VAPOURSYNTH` (10), `VS_PLUGINS` (11), and on an NVIDIA RTX card `VS-MLRT LEAN` (14). The order matters: plugins need the engine already in place.

Decision tree — which preset for which job

Read top-down: the first matching case is your preset.

Material symptom	Preset	Palette
Interlaced legacy (DV, HDV, VHS captures, broadcast TS)	<code>qtgmc_deinterlace</code>	restoration
NTSC 29.97 fps with 3:2 pulldown (telecined film)	<code>tivtc_ivtc</code>	restoration

Material symptom	Preset	Palette
High-ISO / night noise, must keep detail	<code>mvtools_mcdegrain</code>	restoration
Rainbow / dot crawl on VHS-rip or composite capture	<code>derainbow_decross</code>	restoration
Over-compressed H.264/MPEG (YouTube-rip, WhatsApp, SD broadcast)	<code>deblock_h264_artefacts</code>	restoration
Haze / flat low-contrast — need to "lift" the picture	<code>dehaze_local_contrast</code>	restoration
ML upscale 1080p → ~4K (sharp faces, fabric texture)	<code>vsmlrt_realesrgan_2x</code>	restoration
ML interpolation 24 → 60fps (smoother than Optical Flow)	<code>vsmlrt_rife_60fps</code>	restoration
ML denoise of heavy noise / high-ISO without texture loss	<code>dpir_denoise</code>	restoration
Digital noise in shadows only ("night" digital footage)	<code>shadow_denoise_sota</code> ★	precision
Light uniform noise, hardware-agnostic (no CUDA/OpenCL)	<code>mild_denoise</code>	precision
Chroma noise only (dirty blue channel)	<code>chroma_cleanup</code>	precision
Banding in gradients (skies, walls)	<code>deband_safe</code> or <code>deband_fine_grain</code>	precision
One preset for everything filmic	<code>filmic_rebuild</code> ★	precision
Prep for color grading in DaVinci	<code>pregrade_prep</code>	precision
Clean archival master without grain	<code>archive_clean</code>	precision
Subtle filmic look, safe default	<code>cinematic</code>	film_looks
Specific 35mm Kodak (250D / 500T / 50D)	<code>film_35mm</code>	film_looks
16mm organic, lifted shadows	<code>film_16mm</code>	film_looks
Super 8, faded 70s, heavy grain	<code>super8</code>	film_looks
High-contrast desaturated (Se7en, Saving Private Ryan)	<code>bleach_bypass</code>	film_looks
Large-format IMAX feel — minimal grain, gate weave	<code>imax_70mm</code>	film_looks

Material symptom	Preset	Palette
Hollywood cinemascope (horizontal blue lens flares + teal shadows)	<code>anamorphic_scope</code>	film_looks
VHS / CRT aesthetic	<code>vhs_crt</code>	retro
90s amateur camcorder	<code>camcorder_90s</code>	retro
1970s Polaroid SX-70 (candy-bloom, warm cream)	<code>polaroid</code>	retro

Pipeline scenarios (chain multiple presets via `apply-profile-batch` or manual runs):

Scenario	Steps
Night timelapse → film	<code>shadow_denoise_sota</code> → <code>filmic_rebuild</code> → <code>cinematic</code>
VHS archive → restored master	<code>qtgmc_deinterlace</code> → <code>derainbow_decross</code> → <code>mvtools_mcdegrain</code> → <code>archive_clean</code>
Old YouTube-rip → project-ready	<code>deblock_h264_artefacts</code> → <code>dehaze_local_contrast</code> → <code>pregrade_prep</code>
Telecined NTSC DVD → 24p master	<code>tivtc_ivtc</code> → <code>archive_clean</code>
Modern digital → film look	<code>mild_denoise</code> → <code>film_35mm</code>

PRECISION palette — technical pipeline (8 presets)

Menu order = pipeline logic: Stage 1 (denoise) → Stage 2 (deband) → Stage 3 (compositions).

Stage 1 — denoise

`mild_denoise` — soft universal denoiser

DFTTest spectral denoiser. Pure CPU, hardware-agnostic — same behaviour on any CPU.
Recommended as the **first choice** when nothing specific is known about the material: light noise, gentle clean-up without risk.

- Parameters: `strength` = light (sigma=4.0) / medium (8.0) / strong (14.0)
- Encoder: `h264_crf17` or `h264_crf14` for archival quality

- What's in NLE: only Temporal NR / generic Noise Reduction; coarser results

`shadow_denoise_sota` ★ — **flagship shadow denoiser**

BM3D with float32 pipeline + smooth luma-mask "shadows only". Noise is killed **only** in zones below `shadow_threshold`; the rest of the picture is untouched. Chroma planes are always denoised (chroma noise is equally ugly everywhere). Optional `grain_back` re-adds micro-grain so cleaned shadows do not look "plastic".

- Parameters: `sigma=2.5` (BM3D luma), `use_cuda=0/1`, `grain_back=0.6`, `shadow_threshold=0.20`, `transition=0.10`
- On NVIDIA: `--use-cuda 1` gives 25–35% wall-time win on 4K
- Encoder: `h264_crf17` / `prores_lt`
- Not in NLE: zone-targeted denoise with smooth blend, no cheap "threshold mask" hack

`chroma_cleanup` — **chroma-only cleanup**

DFTTest on planes=[1,2]. Luma untouched. Use when "the blue channel is dirty" (typical on old compact cameras, low-bitrate AVCHD).

- Parameters: `strength` = light / medium / strong
- Encoder: `h264_crf17`
- Not in NLE: clean chroma-only DFTTest without luma side-effects

Stage 2 — deband

`deband_safe` — **safe debander**

neo_f3kdb with light settings, no grain. Removes banding in gradients (skies, walls, night lighting) without softening edges.

- Parameters: `range=15`, `y=64`, `cb=64`, `cr=64`
- Encoder: `h264_crf17` / `h265_crf21`

`deband_fine_grain` — **deband + fine-grain restore**

Same neo_f3kdb + AddGrain. Light monochromatic grain after debanding — avoids the "too clean" look characteristic of cheap compression.

- Parameters: `range=15`, `grain_var=1.5`
- Encoder: `prores_lt` (for downstream grading)

Stage 3 — compositions

filmic_rebuild ★ — flagship composition

Full chain: BM3D denoise → neo_f3kdb deband → 3-zone luma grain (shadows / midtones / highlights). Zonal grain — heavier in shadows (like real film), lighter in highlights. Recommended as "one preset for everything filmic".

- Parameters: `sigma=2.0`, `use_cuda=0/1`, `deband_range=14`, `grain_shadow=1.5`, `grain_mid=0.9`, `grain_high=0.4`, `shadow_threshold=0.30`, `high_threshold=0.65`
- Encoder: `prores_lt` / `h265_crf21`

archive_clean — neutral archival master

DFTTest + neo_f3kdb. **No grain**. Goal — maximally clean, "flat" picture for archiving. Do not use if downstream grading is planned (grading works better on grainy material).

- Parameters: `denoise=medium`, `range=15`
- Encoder: `prores_422hq` / `prores_422hq_mxf`

pregrade_prep — handoff to DaVinci

Minimum touch: light DFTTest only. No deband, no grain. Goal — give Resolve the cleanest source so the colorist isn't grading on top of compression artefacts.

- Parameters: `strength=light`
- Encoder: `prores_lt_mxf` (round-trip with Adobe / Avid)

FILM_LOOKS palette — film emulations (7 presets)

Each look uses the shared `audion_lib` library (MTF softening, halation bloom, zone grain, gamma curve, black-lift, desaturation).

cinematic — universal subtle filmic

Safe default. Light MTF softening + halation + micro-grain. Not tied to a specific stock. Works on any modern digital material.

- Parameters: `intensity=1.0` (range 0..2)
- Encoder: `prores_lt` / `h264_crf17`

film_35mm — Kodak 35mm with stock variants

Emulates real Kodak stocks: 250D (daylight balanced), 500T (tungsten), 50D (fine grain). Each stock has its own gamma curve, balance, grain density.

- Parameters: `stock=250D|500T|50D`, `intensity=1.0`
- Encoder: `prores_lt` / `prores_lt_mxf`

film_16mm — organic, grainier

Lifted blacks, more pronounced grain, softer than 35mm. Suits "documentary" / arthouse aesthetics.

- Parameters: `intensity=1.0`
- Encoder: `prores_lt`

super8 — strongest look

Heaviest grain, softest optics, faded 70s. For music videos / stylized inserts.

- Parameters: `intensity=1.0`
- Encoder: `h264_crf17` (grain is already baked in, aggressive compression OK)

imax_70mm — large-format IMAX feel ★

Opposite of Super 8: **minimal grain, maximum resolution**, very soft halation on highlights, optional 1px gate weave (deterministic, seed=42 → reproducible). The huge frame of real 70mm IMAX → each silver grain is tiny relative to the picture. Suits shots that should feel "epic", not "filmic".

- Parameters: `intensity=1.0`, `gate_weave=1` (on by default)
- Encoder: `prores_lt` / `h265_crf17`
- Not in NLE: calibrated micro-grain plus gate weave (NLEs only do flat plate grain)

anamorphic_scope ★ — Hollywood cinemascope

Anamorphic optics signature: **long horizontal blue lens flares** on highlights only (those "horizontal blue streaks across the sky" in Abrams / Nolan / Villeneuve films, made by Panavision / Hawk / ARRI Master Anamorphic). Plus subtle teal-cool shadows for the classic blockbuster look.

In our implementation: highlight-mask (`flare_thr`) → horizontal low-pass blur of length `flare_len` → blue tint (R 10% / G 55% / B 100%) → additive merge over the original. Then shadow-masked teal shift (R -10% / G +5% / B +12%). All in 16-bit RGB so streaks don't clip.

- Parameters: `intensity=1.0`, `flare_len=96` (48 subtle / **96 default** / 160 dramatic / 256 extreme), `flare_thr=0.78` (0.85 highlights-only / **0.78 default** / 0.70 aggressive / 0.60 heavy), `teal_shadow=0.35` (0.0 off / 0.20 subtle / **0.35 default** / 0.60 heavy)
- Encoder: `prores_lt` (for colorist) or `h265_crf14` (final)
- The preset does **not** crop the frame to 2.39:1 — that's a creative choice; add `ffmpeg -vf crop=W:floor(W/2.39):0:Y` in post if you need true scope.
- Not in NLE: directional anamorphic streak without expensive paid plugins like Optical Flares; smooth threshold for flare cut-in; full 16-bit math.

`bleach_bypass` — high-contrast desaturated

"Se7en" / "Saving Private Ryan" look. High contrast, silver greys, blown highlights. Hard stylistic choice.

- Parameters: `intensity=1.0`
- Encoder: `prores_lt`

RETRO palette — analog character (3 presets)

`vhs_crt` — VHS / CRT

Chroma bleed (horizontal color smear), soft optics, analog noise. Emulation of VHS playback through a CRT TV.

- Parameters: `intensity=1.2`
- Encoder: `h264_crf17`

`camcorder_90s` — amateur camcorder

Softer than VHS, slight overexposure, minimal chroma bleed. 90s home video aesthetic.

- Parameters: `intensity=1.0`
- Encoder: `h264_crf17`

polaroid ★ — 1970s Polaroid SX-70

Strong candy-bloom on highlights, lifted blacks (never goes to true black, creamy fade), warm gamma push, mild desaturation, **radial edge vignette**. Five composited steps — what would take a stack of 5–6 effects to build manually in NLE.

- Parameters: `intensity=1.0`, `vignette=0.55` (range 0..1; 0 = none, 1 = corners to black)
- Encoder: `h264_crf17` (filmic material tolerates compression)
- Not in NLE: one-shot Polaroid emulation; radial vignette without geometric distortion or LUT side-effects

RESTORATION palette — supercannons (10 presets, Phase 18.B + 18.B-ML)

What's **missing or weak** in Adobe Premiere / DaVinci Resolve.

qtgmc_deinterlace — reference-grade deinterlace

havsfunc.QTGMC (NNEDI3 + MVTools motion estimation). Reconstructs progressive frames from interlaced source via neural-network upscaling of each field + motion-compensated temporal smoothing. **Better than any NLE out-of-the-box.**

- Parameters: `field_order=tff|bff`, `qtgmc-preset=Faster|Fast|Medium|Slow|Slower|Placebo`, `output_fps=single|double`
- When to use: DV, HDV, VHS captures, broadcast TS, S-VHS / U-matic digitization
- Encoder: `prores_lt` (for downstream work) or `h264_crf17` (if final master)

tivtc_ivtc — inverse telecine

TIVTC.TFM (field matching) + TIVTC.TDecimate (drop duplicate). Reference-quality 3:2 pulldown removal: NTSC 29.97i → 23.976p. Recovers the original 24p film master from a telecined source.

- Parameters: `pp=6` (TFM post-processor), `cycle=5`, `rdrop=1` (standard NTSC pattern)
- When to use: NTSC DVDs, broadcast prints, digitized film transfers
- Encoder: `prores_422` / `prores_422hq_mxf`

mvtools_mcdegrain — motion-compensated denoise

MVTools motion estimation → MDeGrain temporal averaging along motion vectors. **Removes noise WITHOUT losing detail** — what Topaz Video Enhance AI and Neat Video do internally. DaVinci Temporal NR is a coarse implementation of the same; here you get reference-grade.

- Parameters: **radius=2** (5-frame window), **thsad=200**, **blksize=16** (HD) / **8** (4K detail)
- When to use: high-ISO night footage, preserving skin / fabric texture
- Encoder: **prores_lt** / **h265_crf17**

derainbow_decross — NTSC composite cleanup

MVTools-based motion-compensated chroma-only smoothing. Removes rainbow / dot crawl / cross-color on composite-capture material (VHS-rip, U-matic, BetaSP, S-Video → SDI). Luma untouched.

- Parameters: **strength=0.6**, **blksize=16**
- When to use: digitized VHS, color "crawls" on thin lines, mosquito noise on B/W edges
- Encoder: **prores_lt** / **h264_crf17**

deblock_h264_artefacts — over-compression rescue

havsfunc.Deblock_QED — edge-aware deblocker for H.264 / MPEG-2 / MPEG-4. Smooths 8x8 block boundaries, ringing around edges, mosquito noise. Edge-aware — does not smear real detail.

- Parameters: **quant1=24**, **quant2=26**, **aoffset=1**, **boffset=1**
- When to use: old YouTube rips, WhatsApp/Telegram re-encodes, low-bitrate SD broadcast TS
- Encoder: **h264_crf17** (after deblock the material is cleaner; you can re-encode at higher CRF)

dehaze_local_contrast — clarity without halos

Luma-only local contrast via high-pass + zone-weighted MaskedMerge. Zone-mask (parabolic, peak at midtones) prevents blown highlights and crushed shadows. Color does not shift (luma-only). 16-bit math.

- Parameters: **strength=1.0**, **radius=8** (clarity) / **12-16** (haze removal)
- When to use: haze, flat low-contrast material, need to "wake the picture up" without destroying highlights
- Not in NLE: Resolve "Dehaze" blows highlights and shifts color; here, no halos
- Encoder: **prores_lt** (handoff to colorist) or **h264_crf17** (final)

Encoders — my patterns

(Audion default workflow per the 2026-04-28 discussion)

Quality ladder (same 14 / 17 / 21 tiers across software, NVENC, QuickSync, and AMF): **14** → **17** → **21**, three distinguishable steps without overlap.

Profile	When to use
<code>h264_crf14</code> / <code>h265_crf14</code> ★ default	Semi-lossless archive, master for downstream work. Audion default since 2026-04-28.
<code>h264_crf17</code> / <code>h265_crf17</code>	"Almost invisible" lossy, final master
<code>h264_crf21</code> / <code>h265_crf21</code>	Web preview / proxy / preview
<code>prores_lt</code> ★	Audion default ProRes — no keying, made for grading and round-trip
<code>prores_lt_mxf</code> ★	ProRes LT in MXF wrapper — for Adobe Premiere / Avid round-trip
<code>prores_422</code>	If you need slightly more bitrate than LT
<code>prores_422_mxf</code>	ProRes 422 in MXF wrapper for Adobe / Avid handoff
<code>prores_422hq</code>	Only if keying is planned
<code>prores_422hq_mxf</code>	HQ + MXF for broadcast / Avid finishing
<code>dnxhr_lb</code> / <code>_sq</code> / <code>_hq</code> / <code>_hqx</code>	Avid-style DNxHR (LB low / SQ standard / HQ high / HQX 10-bit)
<code>h264_nvenc_q14</code> / <code>_q17</code> / <code>_q21</code>	NVENC hardware H.264 on NVIDIA. Removes the libx264 CPU bottleneck — VS filters don't compete with encode. CQ ≈ CRF . 8-bit yuv420p.
<code>h265_nvenc_q14</code> / <code>_q17</code> / <code>_q21</code>	NVENC hardware HEVC, 10-bit p010le. Ideal for full pipeline on NVIDIA: VS filtering on CPU/CUDA + encode on NVENC = no CPU bottleneck.
<code>h264_qsv_q14</code> / <code>_q17</code> / <code>_q21</code>	Intel QuickSync H.264. Good for Intel iGPU batch/proxy work when CPU should stay free for VapourSynth.
<code>h265_qsv_q14</code> / <code>_q17</code> / <code>_q21</code>	Intel QuickSync HEVC, p010le for 10-bit output where supported.
<code>h264_amf_q14</code> / <code>_q17</code> / <code>_q21</code>	AMD AMF H.264 via CQP, matching the same visual ladder.

Profile	When to use
<code>h265_amf_q14</code> / <code>_q17</code> / <code>_q21</code>	AMD AMF HEVC, p010le output; useful on Radeon hosts.

Rule: for any non-final pass → `prores_lt` / `prores_lt_mxf` or DNxHR if the receiving app prefers it. For final delivery → `h264_crf14` / `h264_crf17` (software, best density per bit) OR the matching hardware ladder for the host (`nvenc`, `qsv`, `amf`) when wall-time matters.

When to use hardware encode vs software

Scenario	Encoder
Heavy VS pipeline (filmic_rebuild, mvtools_mcdegrain) on NVIDIA	NVENC — CPU is freed for the VS filter, total wall-time drops 30-50%
Final delivery where maximum bit density matters	libx264/libx265 CRF — software encoder is still slightly more efficient at low CRF
Intel iGPU host / laptop batch	QuickSync — good throughput with low CPU pressure
AMD/Radeon host	AMF — same 14/17/21 CQP ladder, avoids CPU encode bottlenecks
Big batch over hundreds of files	NVENC / QSV / AMF — time savings compound
No working hardware encoder	Software only (<code>h264_crf*</code> / <code>h265_crf*</code>)

NVENC requires: NVIDIA driver R525+. Quality: NVENC on Ada/Blackwell (RTX 40/50) is **comparable** to libx264 medium at the same CQ; on older generations (Pascal/Turing) software CRF wins slightly on bitrate efficiency for the same quality, but NVENC is still many times faster.

Install & service scripts — when to run what

There are many scripts in `install/`, and not all are obvious. This section is the "when do you call which" map with concrete arguments and examples. Every script comes as a `.cmd` (thin wrapper, resolves portable PowerShell) + `.ps1` (full logic) pair. Usually you launch them via `builder_main.cmd` (FZF menu), but any can be run directly via double-click or CLI.

Entry point: `builder_main.cmd`

Main menu of service operations. FZF navigation (if `system_core\fzf.exe` exists), otherwise CMD fallback with letter hotkeys. Menu map:

#	Item	What it does	Underlying script
[01]	Build portable env CMD builder	initial build of <code>runtime/</code>	<code>Build_Portable_Env_Build.cmd</code>
[02]	Build portable env PS	alt PowerShell-based builder	<code>Build_Portable_Env.ps1</code>
[03]	Install portable offline	offline install from a pre-downloaded wheelhouse	<code>install_portable_offline.cmd</code>
[04]	Verify portable env	orchestrator-Python sanity	<code>verify_portable_env.cmd</code>
[05]	Update FZF	bump <code>fzf.exe</code> to latest	<code>launcher-tools-update_fzf.cmd</code>
[06..08]	Licenses	collect / prune / dedupe license files	<code>system_core\license\Run-*.cmd</code>
[09]	Make release archive	release zip excluding <code>output/</code> , <code>logs/</code> , <code>._runtime/</code> , <code>install/download/</code> , private <code>MEMORY.md</code>	<code>make_release_archive.cmd</code>
[04]	PowerShell	portable pwsh 7 → <code>system_core\powershell\</code>	<code>Install-Portable-PowerShell.cmd</code>
[10]	VapourSynth	latest VS stable + latest Python 3.12.x embed → <code>system_core\vapoursynth\</code>	<code>Install-Portable-VapourSynth.cmd</code>
[11]	VS plugins	vsrepo plugin set (incl. <code>bm3dcuda</code> and <code>znedi3</code> by default)	<code>Install-VS-Plugins.cmd</code>
[12]	FFmpeg	Exact-stable Gyan by driver policy; BtbN rolling opt-in → <code>Tools\ffmpeg\bin\</code>	<code>Install-Portable-FFmpeg.cmd</code>

#	Item	What it does	Underlying script
[13] ★	VS-MLRT LEAN	fresh-download vs-mlrt TensorRT bundle, clean <code>plugins\vsmlrt</code> , lean-trim to current SM	<code>Install-VS-mlrt.cmd /LEAN</code>
[14]	VS-MLRT FULL	fresh-download same install, no trim — for USB distribution	<code>Install-VS-mlrt.cmd /FULL</code>
[70]	Clean install cache	remove transient install downloads, staging dirs, and bytecode caches while preserving portable payloads	<code>Clean-Install-Cache.cmd</code>
[90..99]	Open / Project	explorer for subfolders / jump to project launcher	—
[00]	Exit	—	—

From-scratch order: [01] -> [03] -> [04] -> [10] -> [11] -> [12] -> (optional) [13] -> [70]. After all install steps, run `runtime\python.exe system_core\doctor.py` once — it should be all green.

`install\Install-Portable-VapourSynth.{cmd,ps1}` — latest VS stable + own Python 3.12.x

Resolves the latest stable VapourSynth release and latest Python 3.12.x embed, cleans `system_core\vapoursynth\`, installs the matching VapourSynth wheel, and drops the `portable.vs` marker so vsrepo enters portable mode. After the wheel install it explicitly creates and prints the active plugin dir from `vapoursynth.get_plugin_dir()`; VS R74+ no longer treats legacy `vs-plugins\` as the real autoload target. Uses `Expand-7zArchive` (via `Ensure-7zip.ps1`) — ~3× faster than `Expand-Archive` on large archives.

At the end calls `Repair-PipShims.cmd` — fixes shebangs in `Scripts\*.exe` immediately after wheel install (a fresh install needs no manual fix).

Flags: `/R <rev>` (e.g. `/R R76` to pin version). `/F` is accepted for compatibility; current installer already refreshes by default.

`install\Install-VS-Plugins.{cmd,ps1}` — VapourSynth plugins via vsrepo

Installs:

- v1.0 set: `lsmas`, `ffms2`, `fmtconv`, `neo_f3kdb`, `addgrain`, `knlmeanscl`, `bm3dcpu`, `dfttest`
- Restoration set (Phase 18.B): `havsfunc`, `mvsfunc`, `mvtools`, `tivtc`
- CUDA: `bm3dcuda` **by default** (Phase 18.A0). Opt-out via `/NO-CUDA` (on a clean non-NVIDIA host, to skip a plugin that would fail at load anyway).

Also installs `vsutil` via pip (imported at the top level of `havsfunc.py` — without it, Restoration presets crash with `ModuleNotFoundError`).

Update-safe: `vsrepo update` refreshes the package database, the installer cleans the active `vapoursynth.get_plugin_dir()` path while preserving `plugins\vsmlrt` if present, then `vsrepo install` installs the requested list again. Safe to re-run after updating the plugin list.

`install\Install-Portable-FFmpeg.{cmd,ps1}` — BtbN GPL latest, Gyan.dev fallback

Installs a driver-compatible exact-stable Gyan build by default. A BtbN rolling release-branch build remains available only by explicit opt-in. Drops `ffmpeg.exe` / `ffprobe.exe` / `ffplay.exe` into `Tools\ffmpeg\bin\`. Fast extraction via `Expand-7zArchive`.

Flags: `/V <variant>` (`gpl` default / `lgpl` / `gpl-shared`), `/F` (force).

`install\Install-VS-mlrt.{cmd,ps1}` — ML stack (Phase 18.B-ML)

Only for those who want the ML presets (`vsmlrt_realesrgan_2x`, `vsmlrt_rife_60fps`, `dpir_denoise`). Installs the full TensorRT bundle: VSTRT + VSORT + VSOV + VSNCNN + ONNX runtime + TensorRT runtime DLLs + the complete ONNX model collection. After extraction, Audion moves OpenVINO/vsov out of the active autoload folder to `system_core\vapoursynth\disabled_plugins\vsmlrt-openvino`: TensorRT / TensorRT-RTX / ONNX Runtime remain active, while Windows `Bad Image 0xc0e90002` from OpenVINO DLLs cannot break startup.

Working set after install: ~3.5 GB at `/LEAN` (default), ~10 GB at `/FULL`. Every run fresh-downloads the selected vs-mlrt assets and cleans only the `plugins\vsmlrt` subtree before copying runtime DLLs, scripts and models. Extraction via `7zr.exe` (needed for BCJ2-filtered streams in TRT runtime DLLs — `py7zr` cannot decode them).

Full update order: if you run `[10] VAPOURSYNTH` or `[11] VS PLUGINS`, run `[13] VS-MLRT LEAN` again before any Full all-presets smoke. ML presets should never be validated against an older MLRT layer after the base VS runtime/plugins were refreshed.

Flags:

- `/LEAN` (default via `[13]`) — trims TensorRT builder resources to the current SM (detected via `nvidia-smi --query-gpu=compute_cap`), removes unused models (cugan, waifu2x). On non-NVIDIA, falls back to FULL behaviour.
- `/FULL` (via `[14]`) — full bundle for USB distribution to another machine.
- `/DROP-CACHE` — after success, calls the central `Clean-Install-Cache` policy for transient install downloads, staging dirs, and bytecode caches.
- Default from builder = `/LEAN` and keeps install cache. Use `[70] Clean install cache` or `/DROP-CACHE` explicitly to remove cached downloads.

After install, the backend in ML presets is selected **automatically**: TRT (if NVIDIA + matching TRT runtime) → ORT_DML (DirectML, any DX12 GPU including Intel Xe / AMF) → ORT_CPU. Cross-vendor stable (via `audion_lib.vsm_lrt_backend_chain()`) — on non-NVIDIA we don't even attempt to compile a TRT engine, saving 30–120 s).

Annoyance: Windows Defender blocks the unsigned `openvino_intel_npu_plugin.dll`. The script removes it automatically after extract (not needed on NVIDIA / DirectML setups).

`install\Clean-Install-Cache.{cmd,ps1}` — disk reclaim

Removes transient install downloads (`.7z`, `.zip`, `.tar.*`, `.msi`, `.exe`), exact installer staging dirs, and Python bytecode caches outside payload/user-data zones. Preserve list inside `install\download\` (always kept):

- `.gitkeep`
- `get-pip.py` (re-used on every fresh build)
- `7z*-extra.7z` (Ensure-7zip bootstrap helper, ~2 MB)

When you need it: after a successful `[13] VS-MLRT LEAN` or `[14] VS-MLRT FULL` (~3.5 GB compressed archives), periodically after `Install-Portable-*` (fresh FFmpeg build / new VS R75+).

Philosophy: **storage > bandwidth**. Re-download = ~1.5 minutes on gigabit; 3.5 GB on disk sit there forever. Working set after lean+cleanup: ~3.5 GB instead of 10.

`install\Repair-PipShims.{cmd,ps1}` — fix `Scripts\*.exe` after a move

Symptom: after moving the project to a different drive letter / path / machine, `vspipe.exe` and other `Scripts\*.exe` silently exit 1 with no output. `doctor.py` shows `[FAIL] vspipe runs` while `python.exe -c "import vapoursynth"` works fine.

Cause: pip embeds an absolute shebang (`#!/<full python.exe path>`) inside each `Scripts\*.exe`. After the move, that path no longer exists.

What to do: run `install\Repair-PipShims.cmd`. It rewrites the shebang in every `Scripts\*.exe` to the current `system_core\vapoursynth\python.exe`. The launcher stub and trailing zip are left intact.

Flags:

- `/WHATIF` or `/N` — dry-run preview without writing.

In normal operation **not needed:** `system_core\engine\selfheal.py` is hooked into the start of `main.py` and `doctor.py` — it reads the first 16 KB of `vspipe.exe` on startup, parses the embedded shebang, and on mismatch silently calls `Repair-PipShims.cmd`. Idempotent via `AUDION_SELFHEAL_DONE=1`. Manual repair is needed only if you hit vspipe outside `main.py` / `doctor.py` (e.g. a direct CLI bench script that doesn't go through self-heal).

`install\Bench-CUDA.{cmd,ps1}` — pure-pipeline CPU vs CUDA on BM3D

What it benches: two BM3D-heavy presets (`shadow_denoise_sota` and `filmic_rebuild`), each run twice — on CPU and on CUDA. Pipeline `vspipe → ffmpeg -f null` (no encoding). No-encode is needed so libx264 doesn't eat all wall-time on a fast multi-core CPU and hide the CUDA win.

How to use:

```
:: Drag-n-drop a file onto Bench-CUDA.cmd, or:
install\Bench-CUDA.cmd "C:\clips\test.mov"
install\Bench-CUDA.cmd "C:\clips\test.mov" 2.5    :: second arg = sigma (default 2.5)
```

Output: 4 timing lines + a summary table CPU/CUDA + Δ %. Nothing is written to disk.

How to read it:

- On the reference DCI 4K ProRes 25-second clip (RTX 5070 / Ryzen 9 5900X):
`shadow_denoise_sota` = -35%, `filmic_rebuild` = -24%.

- On clips shorter than ~5 s the CUDA setup cost is not amortized → result can be noise or even slightly slower than CPU.
- **Task Manager lies:** even on a working CUDA path it often shows 5–10% GPU. BM3D is bandwidth-bound and runs in bursts; surrounding `fmtc` bit-depth conversions and frame-prop tags happen on the CPU. The authoritative "CUDA is alive" signal is the wall-time delta and `[OK] bm3dcuda runtime - live BM3D CUDA invocation succeeded` in `doctor.py`.

`install\Bench-AllPresets.{cmd,ps1}` ★ — smoke across all 28 presets

Walker over `system_core\presets\<palette>*.vpy`. Every preset is run through `vspipe -c y4m --end <Frames-1> | ffmpeg -f null` — pure pipeline, no disk writes. Smoke goal: confirm that (a) the preset parses, (b) every plugin namespace it needs resolves, (c) frames make it to ffmpeg without exceptions.

How to use:

```
:: Drag-n-drop, or:
install\Bench-AllPresets.cmd "C:\clips\test.mp4"
install\Bench-AllPresets.cmd "C:\clips\test.mp4" 30 sweep auto
```

Current local references:

- Portable CPU/ORT fallback: `28/28 PASS` with `Frames=1`, `Cuda=off`, `MLBackend=ort_cpu` (2026-05-16).
- RTX 5070 validation: `36/36 PASS` with `Frames=1`, `Cuda=sweep`, `MLBackend=auto` (2026-05-27). The 36 rows are 28 presets plus the extra CPU/CUDA Precision sweep.

Arguments:

1. Video path (required)
2. Frames per preset (default 30)
3. Cuda mode: `off` / `on` / `sweep` (default off)
 - `off` — every preset on CPU
 - `on` — every Precision preset with `AUDION_VS_USE_CUDA=1`
 - `sweep` — Precision runs twice (CPU and CUDA), other palettes once
4. ML backend: `auto` / `trt` / `ort_dml` / `ort_cpu` / `sweep` (default auto)
 - `auto` — vs-mlrt picks TRT → ORT_DML → ORT_CPU itself

- `sweep` — ML presets run twice (auto and ort_cpu) for GPU vs CPU baseline comparison

How many rows you get:

- `off auto` — one row per preset (28 total)
- `sweep auto` — Precision×2 + others×1 = 8×2 + 7 + 3 + 10 = 36 rows
- `sweep sweep` — Precision×2 + ML restoration×2 + others×1 = 16 + 7 + 3 + 6 + 4×2 = 40 rows

At the end — Summary `Total / PASS / FAIL`; for `-Cuda sweep` a `CPU vs CUDA Δ %` table over Precision presets, plus a JSON report at `logs\bench_all_presets_<TS>.json`.

When to run it:

- After `Install-VS-Plugins` or `Install-VS-mlrt` — confirm every preset is alive (`PASS == Total`).
- After moving the project (post-`Repair-PipShims`).
- After an NVIDIA driver upgrade — to confirm TRT engines rebuild without errors.
- When adding a new preset — make sure it didn't break neighbour imports.

Hidden-but-important details:

- Per-preset env overrides (`$presetOverrides` table in `Bench-AllPresets.ps1`) — without them `AUDION_VS_STRENGTH=medium` (a Precision tag) breaks Restoration presets (`dehaze`, `derainbow`) that parse STRENGTH as float. And `AUDION_VS_MODEL` differs across ESRGAN / RIFE / DPIR.
- Separate stderr temp files for vspipe and ffmpeg (a single cmd.exe pipe cannot `2>file.log` from both processes — `ERROR_SHARING_VIOLATION`).
- Local row vars are **not** named `$cuda` — that would clobber the script param `$Cuda` (PowerShell variables are case-insensitive → `ValidateSet` breaks on assignment).

`install\Ensure-7zip.ps1` — dot-source helper for 7-Zip

Not run directly — other `*.ps1` files dot-source it:

```
. "$PSScriptRoot\Ensure-7zip.ps1"
$exe = Ensure-7zr -ProjectRoot $ProjectRoot
Expand-7zArchive -Archive $zip -Destination $dst -ProjectRoot $ProjectRoot
```

Build env steps `[01]/[02]` explicitly install `7zr.exe` (~1.5 MB, pure 7z extractor with BCJ2 support) and `7za.exe` (~1.7 MB, universal — zip / 7z / tar / gz / bz2 / xz) into `system_core\7zip\` before

Python setup. Fetched once, then travels with the project.

Why we need it:

- `Expand-Archive` (.NET ZipArchive) chokes on ZIPs > 2 GB (BtbN FFmpeg, VS portable) and is memory-hungry.
- `py7zr` cannot decode BCJ2-filtered streams (vs-mlrt TensorRT runtime DLLs).
- `7za` is format-stable and portable.

Quick "when do I call what" map

Scenario	Scripts in order
Fresh install from scratch	builder_main → <code>[01]</code> → <code>[10]</code> → <code>[11]</code> → <code>[12]</code> → <code>[13]</code> → opt. <code>[14]</code> → <code>[16]</code> → doctor.py
Moved the project to another drive / machine	doctor.py (selfheal calls Repair-PipShims itself) — or manually <code>install\Repair-PipShims.cmd</code>
Need/update the ML stack	builder_main → <code>[13] VS-MLRT LEAN</code> after <code>[10]</code> / <code>[11]</code> / <code>[12]</code> , then Bench-AllPresets with <code>-MLBackend sweep</code>
Want to confirm CUDA is actually alive	<code>install\Bench-CUDA.cmd <video></code> — measures real wall-time delta
Want to confirm all 28 presets work locally without NVIDIA	<code>system_core\powershell\pwsh.exe -NoLogo -NoProfile -ExecutionPolicy Bypass -File install\Bench-AllPresets.ps1 -ProjectRoot . -InputFile <video> -Frames 1 -Cuda off -MLBackend ort_cpu</code>
RTX / TensorRT validation	<code>system_core\powershell\pwsh.exe -NoLogo -NoProfile -ExecutionPolicy Bypass -File install\Bench-AllPresets.ps1 -ProjectRoot . -InputFile <video> -Frames 1 -Cuda sweep -MLBackend auto</code>
Disk is tight	builder_main → <code>[16] CLEAN INSTALL CACHE</code> — frees ~3.5 GB compressed archives
Updated NVIDIA driver / swapped GPU	doctor.py (live <code>bm3dcuda</code> smoke) → Bench-CUDA → Bench-AllPresets (TRT engines rebuild on first ML preset)
Added a new preset	Bench-AllPresets for smoke + register in <code>engine/presets.py</code> + add a launcher entry + update this wiki

Scenario	Scripts in order
Preparing release archive	builder_main → [09] Make release archive — output excludes output/, logs/, .runtime/, install/download/, release/, API keys, and private MEMORY.md; MEMORY.example.md stays public

Additional — full reference

- English docstring per preset: at the top of each `system_core/presets/<palette>/<preset>.vpy`
- CLI flags: `runtime\python.exe system_core\main.py run --help`
- Profiles (cross-palette combinations): `runtime\python.exe system_core\main.py list-profiles`
- Recursive batch with mirror folder structure: `apply-profile-batch --recursive`
- Stack health: `runtime\python.exe system_core\doctor.py`
- AI / ML stack install and scenarios: see *Install & service scripts* section above → `Install-VS-mlrt.cmd`

AI / ML (Phase 18.B-ML) — vs-mlrt stack

These presets use **vs-mlrt** (ONNX inference inside VapourSynth). Backend is auto-selected: `trt` (TensorRT, NVIDIA, fastest) → `ort_dml` (DirectML, any DX12 GPU including Intel Xe / AMF) → `ort_cpu` (CPU fallback). Install via `builder_main.cmd` → `[13] VS-MLRT LEAN` (~3.5 GB download, ~6 GB extracted).

`vsmlrt_realesrgan_2x` ★ — ML 2x upscale

Real-ESRGAN — the de-facto reference for ML video super-resolution. On 1080p input it produces ~3840×2160 with face/fabric sharpness that's **unreachable** via Lanczos / Spline36 / Resolve SuperScale.

- Parameters: `model` (general-x4v3 default / animevideov3 / general-wdn-x4v3 / animejanaiV2-L1/L2/L3), `tile=384`, `tile_pad=16`, `backend=auto`
- Encoder: `prores_lt` (for downstream work) or `h265_crf14` (final)
- Not in NLE: real ML upscale (NLE plugins are usually paid add-ons)

- First TensorRT pass compiles an engine in 30-120s (cached after that)

soft_hd_rebuild_2x — cleanup + ML reconstruction for soft HD

For formally-HD footage whose real detail is closer to 480-720p because of soft optics, binning, aggressive OLPF, old codecs, or archive-generation loss. It cleans the frame first, runs Real-ESRGAN 2x, then applies a conservative luma detail pass.

- Parameters: `rebuild=conservative|balanced|aggressive`, `cleanup=light|medium|strong`, plus Real-ESRGAN `model`, `tile`, `tile_pad`, `backend`
- Encoder: `prores_lt` for grading/handoff or `h265_crf14` for final delivery
- When to use: soft 1080p, archive HD with low real frequency content, under-sampled consumer footage

vsmlrt_rife_60fps ★ — ML frame interpolation

RIFE 4.x — modern ML model for frame interpolation. Understands content (not just pixel motion) → handles occlusion, transparent objects, fades correctly. Visibly smoother than Resolve Optical Flow on complex motion.

- Parameters: `fps_mul=2.5` (24→60 default; alternatives 2/3/4), `model=rife_v4.6` (default; v4.4 / v4.9), `backend=auto`
- Encoder: `h264_crf17` or `prores_lt`
- When to use: 24fps film material → 60Hz target, or slow-mo from regular 24/30/60fps footage

dpir_denoise — ML denoise for heavy noise

DPIR — a deep network for denoise that **preserves** skin and fabric texture where BM3D / DFTTest start to smear. Use it when BM3D "kills" detail on high-ISO material.

- Parameters: `strength=10` (sigma 1-50; 5 light / 10 medium / 15 heavy / 25 very heavy / 50 extreme), `model=drunet_color` (default; gray for B/W; deblocking_color for JPEG/MPEG block artefacts), `tile=384`, `backend=auto`
- Encoder: `prores_lt` (handoff to colorist) or `h264_crf14` (final)
- When to use: ISO 6400+, low-light phone, broken-sensor archive

Last updated: 2026-05-27 — 28 presets across 4 palettes, Soft HD Rebuild 2X in Restoration; local CPU/ORT smoke `28/28 PASS` (`Frames=1`, `Cuda=off`, `MLBackend=ort_cpu`); RTX 5070 CUDA/auto smoke `36/36 PASS` (`Frames=1`, `Cuda=sweep`, `MLBackend=auto`).